

PRANVEER SINGH INSTITUTE OF TECHNOLOGY

B.Tech Computer Science and Engineering — Semester Examination Simulation

SUBJECT: DESIGN ANALYSIS AND ALGORITHM

Paper Set:	Set 01	Duration:	3 Hours
Maximum Marks:	100 Marks	Date:	September 2026

SECTION A

Answer all questions in brief. (10 x 2 = 20 Marks)

- Q1(a).** Define Heap Sort and state its key application in computer systems. [2M]
- Q1(b).** What is the primary significance of Matrix Multiplication? Explain in brief. [2M]
- Q1(c).** State the fundamental principles governing Merge Sort. [2M]
- Q1(d).** Write down the basic properties or conditions required for Quick Sort. [2M]
- Q1(e).** Differentiate in brief between Convex Hull and Searching and Divide and Conquer with Examples Such as Sorting. [2M]
- Q1(f).** List any two advantages and limitations of Divide and Conquer with Examples Such as Sorting. [2M]
- Q1(g).** Give a real-world example illustrating Sorting and Order Statistics - Shell Sort. [2M]
- Q1(h).** Discuss the key methodologies and implementations of Heap Sort. [2M]
- Q1(i).** Discuss the key methodologies and implementations of Matrix Multiplication. [2M]
- Q1(j).** Discuss the key methodologies and implementations of Merge Sort. [2M]

SECTION B

Attempt any three of the following questions. (3 x 10 = 30 Marks)

- Q2.** Explain Quick Sort in detail with suitable architectural diagrams and examples. [10M]
- Q3.** Discuss the algorithmic approach for Convex Hull and Searching. Analyze its time and space complexity. [10M]
- Q4.** Compare and contrast Divide and Conquer with Examples Such as Sorting with Sorting and Order Statistics - Shell Sort. Highlight performance trade-offs. [10M]
- Q5.** Design a solution using Sorting and Order Statistics - Shell Sort for an enterprise system and trace each step. [10M]
- Q6.** Explain the theoretical foundation of Heap Sort and prove its correctness. [10M]

SECTION C

Attempt any five questions from this section. (5 x 10 = 50 Marks)

- Q5.** Explain Matrix Multiplication in detail with suitable architectural diagrams and examples. [10M]
- Q6.** Discuss the algorithmic approach for Merge Sort. Analyze its time and space complexity. [10M]
- Q7.** Compare and contrast Quick Sort with Convex Hull and Searching. Highlight performance trade-offs. [10M]

- Q8.** Design a solution using Convex Hull and Searching for an enterprise system and trace each step. **[10M]**
- Q9.** Explain the theoretical foundation of Divide and Conquer with Examples Such as Sorting and prove its correctness. **[10M]**
- Q10.** Write a step-by-step algorithm or procedure for Sorting and Order Statistics - Shell Sort with illustrative inputs. **[10M]**
- Q11.** Discuss the various challenges encountered in Heap Sort and how they are mitigated. **[10M]**